Mini Project 2 - Usability Report

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NYU CS-GY 6313 Information Visualization

I. Testing the Dashboard

After reviewing the feedback provided along with the submission of my video recording and testing the dashboard prototype with two peers, the main concern seemed to be regarding visual clarity.

The first bit of feedback I received was regarding the choropleth maps, where their representation was a bit difficult to determine due to the lack of labels. In order to make this more clear, I added a legend to show to display the scale of the colors, as well as a title that felt more intuitive, as shown below with one of the maps:

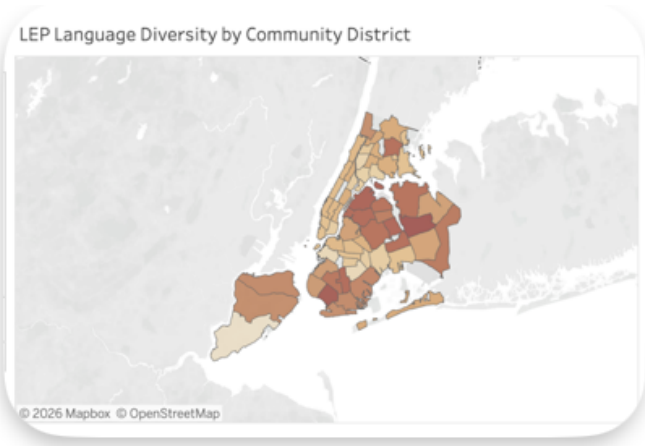

Fig. 1. Before Labeling

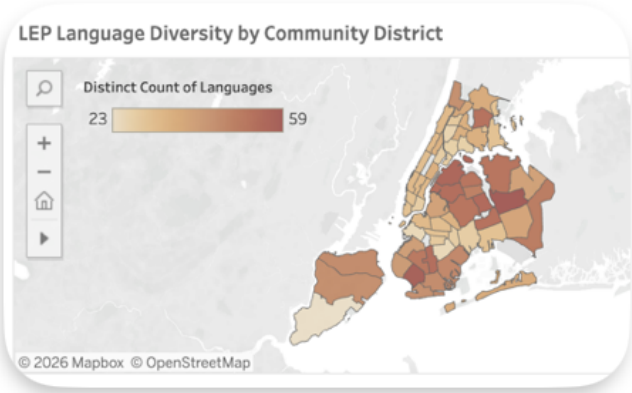

Fig. 2. After Labeling

After adjusting the choropleth maps, there are a few toggles in the dashboard that I was mostly concerned about being intuitive to use for users, so the first two tasks I asked for peers were toggling the number of displayed languages and using the brushing to highlight or exclude certain boroughs. When observing my peers using the dashboard, I noticed that they were having a difficult time discerning which view each toggle corresponds to, due to both proximity and unclear labels. Although they were successfully able to adjust the number of languages shown and use the expand/collapse toggle, it was clear the placement should be adjusted. One peer mentioned moving the toggle box closer towards the views that they correspond to would be much more helpful to understand. In addition to moving the toggles beside their corresponding views, I adjusted the color pallette to create a similarity grouping between the two bar charts for increased clarity and added a border around the toggle box.

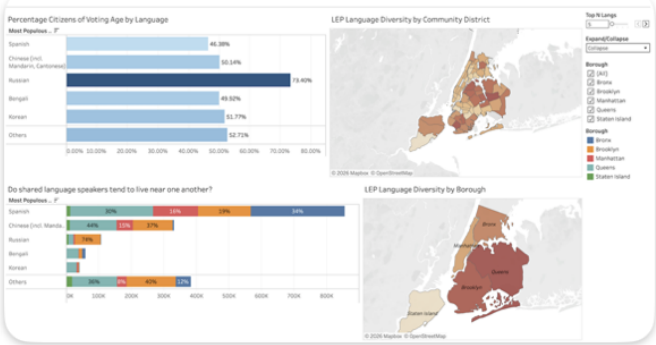

Fig. 3. Before Toggle Realignment

Additionally, when testing the brushing between boroughs and filtering, both peers immediately attempted to do so by clicking directly on the choropleth maps. Using the map directly for highlighting the areas worked, though they both seemed to be confused by a highlighting issue, where the upper left bar chart would be excluded due to it not adjusting for the borough. I did include a mechanism for adjusting the bar chart if you exclude a certain borough, though this required users to go to the toggle box and deselect from a menu, which did not seem to be intuitive after observation. So, I adjusted the dashboard to use the borough selected from the map as a filter that would interact with the bar chart. Additionally, one peer mentioned that it would be helpful to have a place where you can clearly see which borough's data selection you're currently viewing, so I adjusted the legend to highlight which borough is currently selected for clarity.

Another task I wanted to test was users being able to use

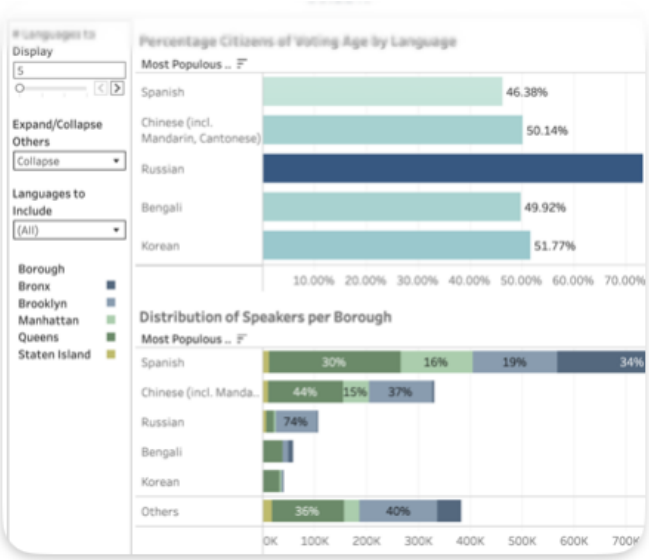

Fig. 4. After Toggle Realignment

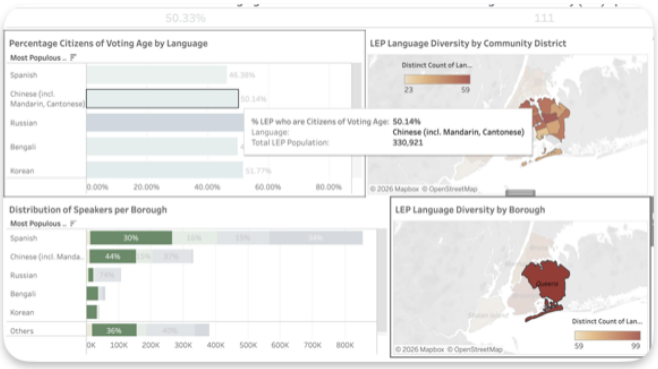

Fig. 5. Before Borough Filter Adjustment

the borough distribution bar chart effectively. Because of the Spanish language's predominance in size, it tends to affect the scaling, so I wanted to see if peers would be able to exclude certain languages from the chart to get a closer view at less populous languages. This caused quite a bit of confusion since there wasn't a toggle for language exclusion, and users needed to click on a language directly from the chart and press exclude, which isn't easily doable for those unfamiliar with Tableau. I also noticed while peers were completing this task that the brushing wasn't applying to both bar charts simultaneously. So, if a certain language was selected, it wouldn't highlight the other chart. Similarly, if a language was excluded from one, it would remain in the other. While trying to complete the exclusion task, one peer mentioned having a selection toggle towards the side for easier use. After thinking on how I might be able to include this without overcrowding the screen, I decided on using a drop-down toggle that is integrated with the number of languages currently displayed to keep the dashboard cleaner. I've included a screenshot of the dashboard with the exclusion toggle and highlight brushing

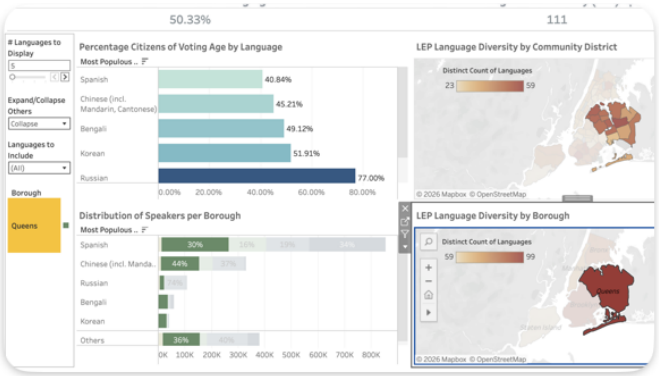

Fig. 6. After Borough Filter Adjustment

below.

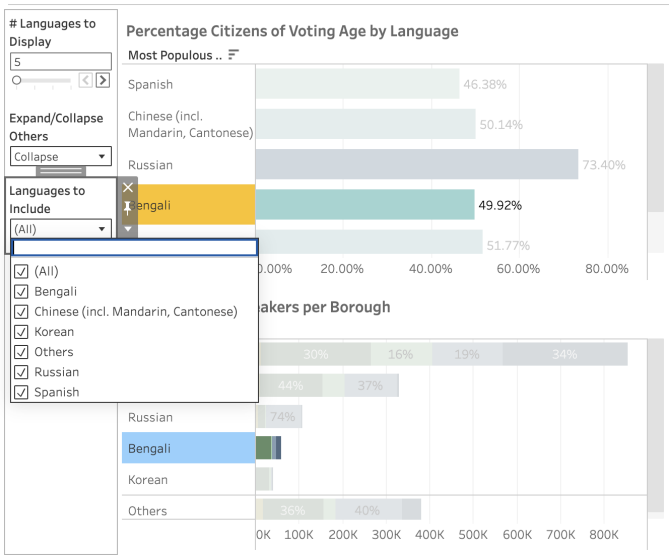

Fig. 7. After Language Brushing and Exclusion Toggle

II. General Feedback

I received a few other general feedback comments, one being that along with the toggle and filter titles being vague, a few acronyms that I had in graph titles should be defined in the dashboard heading metrics. So, I adjusted several of the titles and labels to be more easily understood by those unfamiliar with the data, as well as edited the formatting of the graph titles to appear more substantial for visual clarity. In a similar vein to modifying the headings for clarity, I also decided to keep charts that are in the same vertical plane at the same width, rather than staggered, again for a better visual hierarchy and clarity.

One peer also suggesting adding languages as a hover detail for the choropleth maps. Since the maps are describing the number of languages spoken in that area, they were wishing they could click on an area and see a breakdown of the top languages spoken in that region. I thought this was a wonderful

idea, so I added a small chart that provides the top number of languages in that region, along with the percentage that each language represents from the whole Limited English Proficiency population of the area. The chart also updates with the toggling of the number of languages displayed to keep the view clean and integrated with the rest of the dashboard.

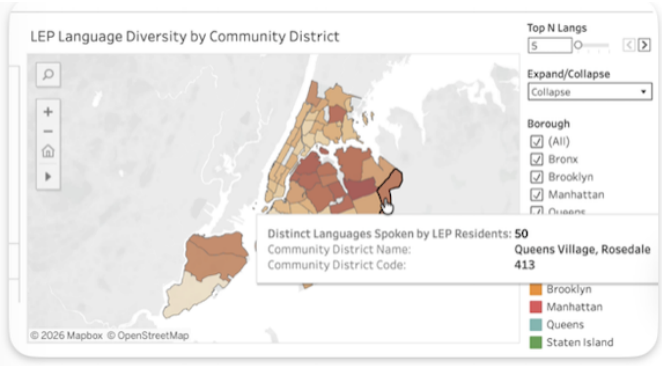

Fig. 8. Before

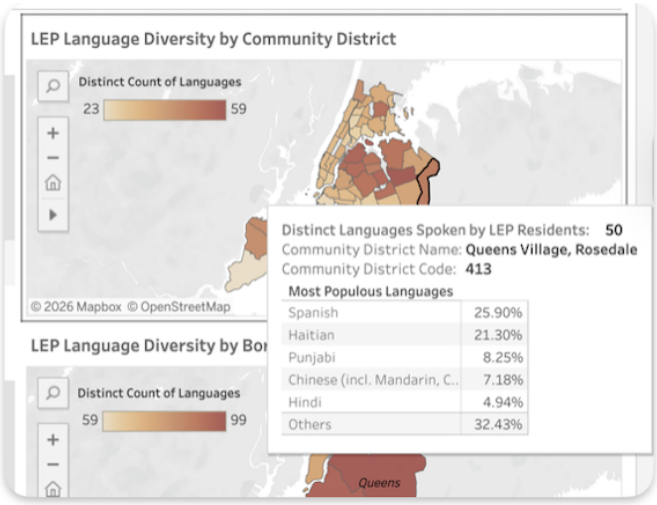

Fig. 9. After